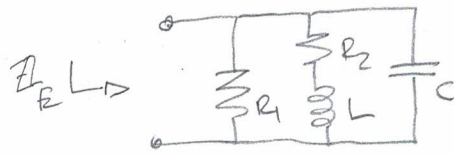


4.47)

BESTÄM EKV. IMPEDANS TILL



$$R_1 = 800 \Omega$$

$$R_2 = 500 \Omega$$

$$L = 200 \text{ mH}$$

$$C = 70 \text{ nF}$$

$$\omega = 4000 \text{ rad/s}$$

Lösning: Alla impedanser är parallelskopplade

vi får

$$\frac{1}{Z_E} = \frac{1}{R_1} + \frac{1}{R_2 + j\omega L} + j\omega C$$

med siffror

$$\left\{ \begin{array}{l} \frac{1}{R_1} = \frac{1}{800} \\ \frac{1}{R_2 + j\omega L} = \frac{1}{500 + j800} \\ j\omega C = j0.28 \cdot 10^{-3} \end{array} \right.$$

vi får

$$\begin{aligned} \frac{1}{Z_E} &= \frac{1}{800} + \frac{1}{500 + j800} + j0.28 \cdot 10^{-3} = \\ &= \frac{1300 + j800}{4 \cdot 10^5 + j6.4 \cdot 10^5} + j0.28 \cdot 10^{-3} = \\ &= \frac{1300 + j800 + j112 - 179.2}{4 \cdot 10^5 + j6.4 \cdot 10^5} = \\ &= \frac{1120.8 + j912}{4 \cdot 10^5 + j6.4 \cdot 10^5} \\ \Rightarrow Z_E &= \frac{4 \cdot 10^5 + j6.4 \cdot 10^5}{1120.8 + j912} = \frac{7.547 \cdot 10^5 \angle 57.99^\circ}{1445 \angle 39.14^\circ} = \\ &= 522.28 \angle 18.85^\circ \end{aligned}$$

SVAR:

$$Z_E = 522.3 \angle 18.9^\circ$$